Project Report: Customer Churn Analysis & Prediction

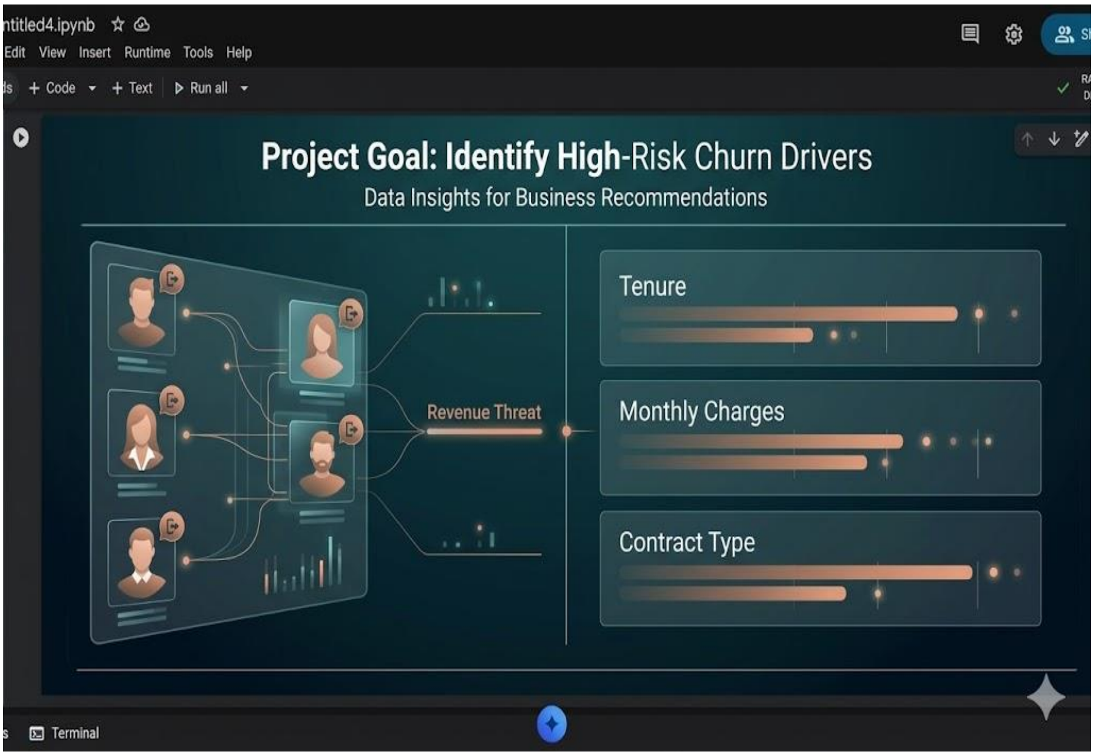
1. Problem Statement:

- Business Problem: Customer attrition (churn) is a major revenue threat.
- Goal: Identify customers likely to leave the service and provide actionable business recommendations to improve retention rates. The goal of this project is to identify customers at high risk of leaving the telecommunications service (churning) and understand the key drivers behind their decision to leave. By predicting churn, the business can proactively engage customers with retention strategies, thereby reducing revenue loss and increasing customer lifetime value.

Data Cleaning: Converted TotalCharges to a numeric format.

- Handled missing values by dropping rows with null entries in the TotalCharges column to maintain data integrity.

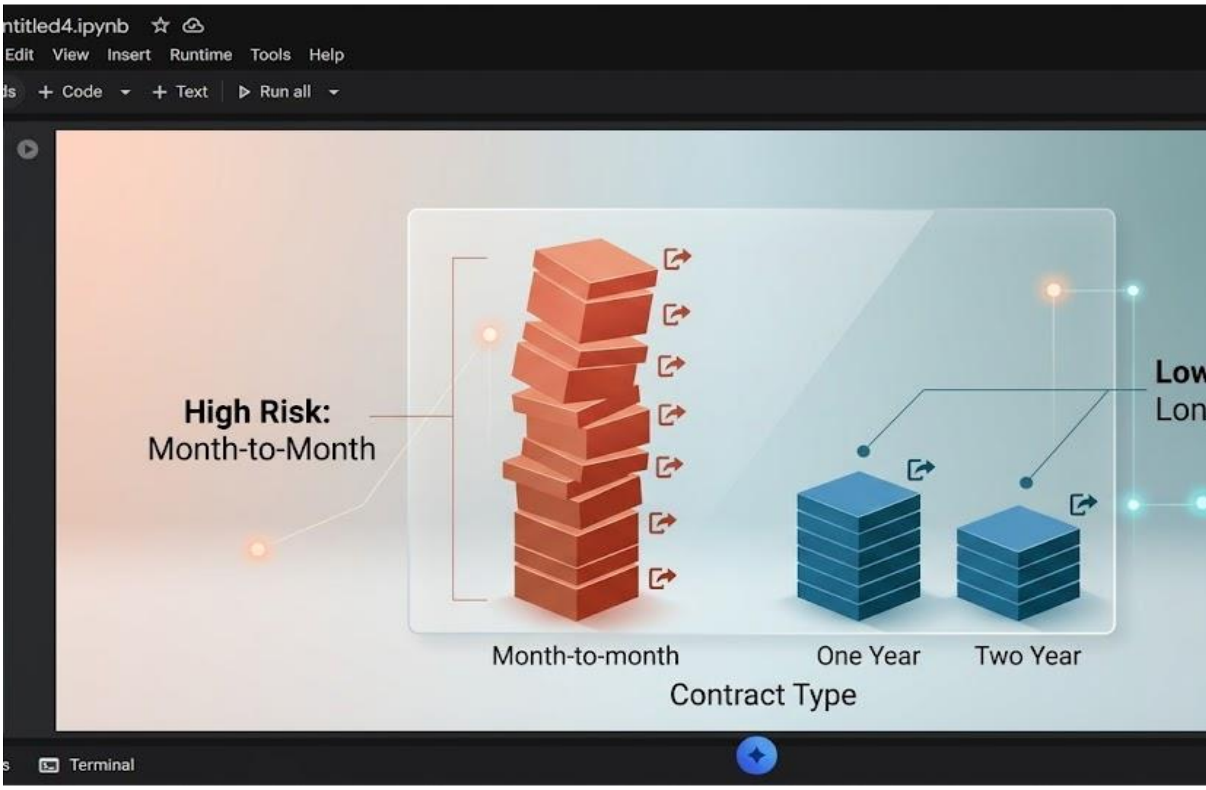
- Encoded the target variable Churn into a binary format ($0 = Stayed$, $1 = Left$).



2. Exploratory Data Analysis (EDA) & Cleaning

- **Data Cleaning:** Handled missing values in TotalCharges and converted categorical text (Yes/No) into binary numerical values for modeling.

- **Patterns Identified:**
    - **Contract Type:** Customers on "Month-to-month" plans have significantly higher churn compared to one- or two-year contracts.
    - **Internet Service:** Fiber optic users are churning at a higher rate than DSL users, suggesting a gap in service value or pricing.



3. Results & Model Evaluation

The models were evaluated based on their ability to correctly identify churners:

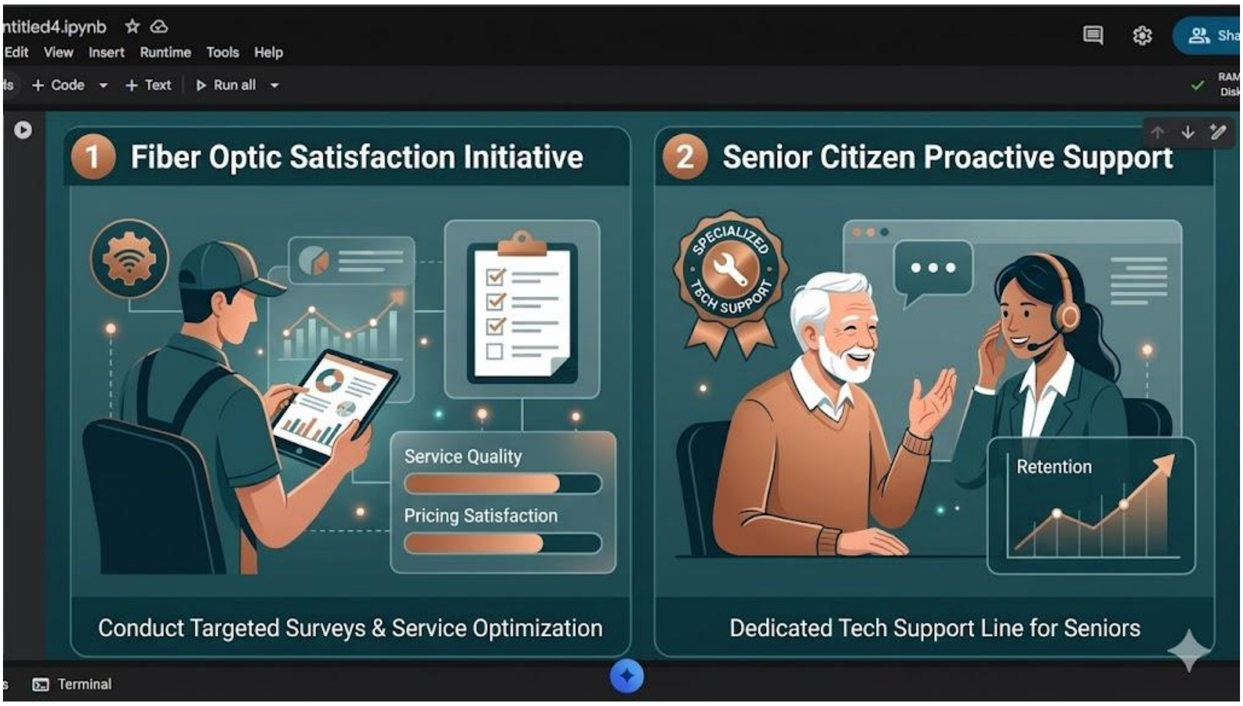
Metric	Logistic Regression	Random Forest
Accuracy	~78%	~77.5%
Recall (Churners)	~0.52	~0.49
Precision (Churners)	~0.62	~0.60

Model Insight: While Logistic Regression provided slightly higher accuracy, both models showed a need for further tuning to improve the "Recall" (identifying all potential churners).

4. Key Influencing Factors

Based on the Random Forest Feature Importance, the top 5 factors influencing a customer's decision to leave are:

1. **Total Charges:** Cumulative spend is a high indicator of churn risk.
2. **Tenure:** Newer customers are significantly more likely to churn than long-term customers.
3. **Monthly Charges:** Higher monthly bills correlate with higher churn rates.
4. **Contract (Month-to-month):** Customers without long-term commitments leave at much higher rates.
5. **Internet Service (Fiber Optic):** Interestingly, fiber optic users showed a higher tendency to churn, suggesting potential issues with service cost or reliability in that segment.



5. Business Insights & Recommendations

- **Target Month-to-Month Contracts:** High churn is concentrated in customers with month-to-month contracts. **Recommendation:** Offer incentives (e.g., a 10% discount) for customers to switch to 1-year or 2-year contracts.
- **Focus on New Customers:** Churn risk is highest in the first few months of service. **Recommendation:** Implement a "New Customer Success" program with check-in calls or loyalty rewards during the first 6 months.
- **Address Fiber Optic Dissatisfaction:** Fiber optic customers are leaving despite the high-speed service. **Recommendation:** Conduct a survey specifically for fiber optic users to identify if the issue is price-related or due to technical instability.
- **Senior Citizen Support:** The data shows Senior Citizens have higher churn rates when facing technical hurdles. **Recommendation:** Provide a dedicated tech-support line for seniors to improve their experience.